

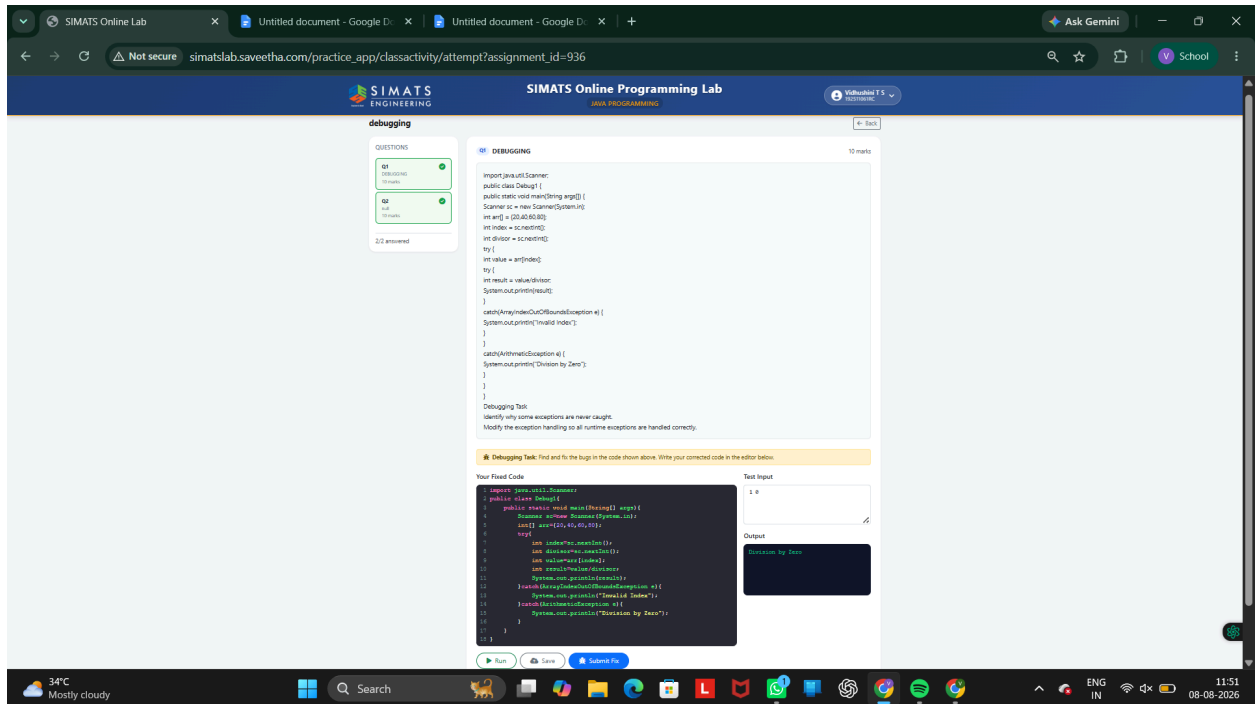
CSA 0921 - Programming in Java

Debugging Program

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1.



The screenshot shows the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" and the sub-header is "JAVA PROGRAMMING". The user is logged in as "Vidhushini T.S". The page is titled "debugging" and shows a "QUESTIONS" sidebar with two questions: "Q1 debugging 10 marks" and "Q2 null 10 marks". The main content area displays a "DEBUGGING" task with a description: "Identify why some exceptions are never caught. Modify the exception handling to all runtime exceptions are handled correctly." The code provided is a Java program that calculates the average of two numbers. The test input is 10 and 20, and the output is "Division by Zero".

```
import java.util.Scanner;

public class Debug4 {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int a[] = {20,40,60,80};
        int index = sc.nextInt();
        int divisor = sc.nextInt();
        try {
            int value = a[index];
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Invalid index");
        }
        try {
            int result = value/divisor;
            System.out.println(result);
        } catch (ArithmeticException e) {
            System.out.println("Division by Zero");
        }
    }
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
import java.util.Scanner;

public class Debug4 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a[] = {10, 40, 60, 80};
        try {
            int index = sc.nextInt();
            int divisor = sc.nextInt();
            int value = a[index];
            int result = value/divisor;
            System.out.println(result);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Invalid index");
        } catch (ArithmeticException e) {
            System.out.println("Division by Zero");
        }
    }
}
```

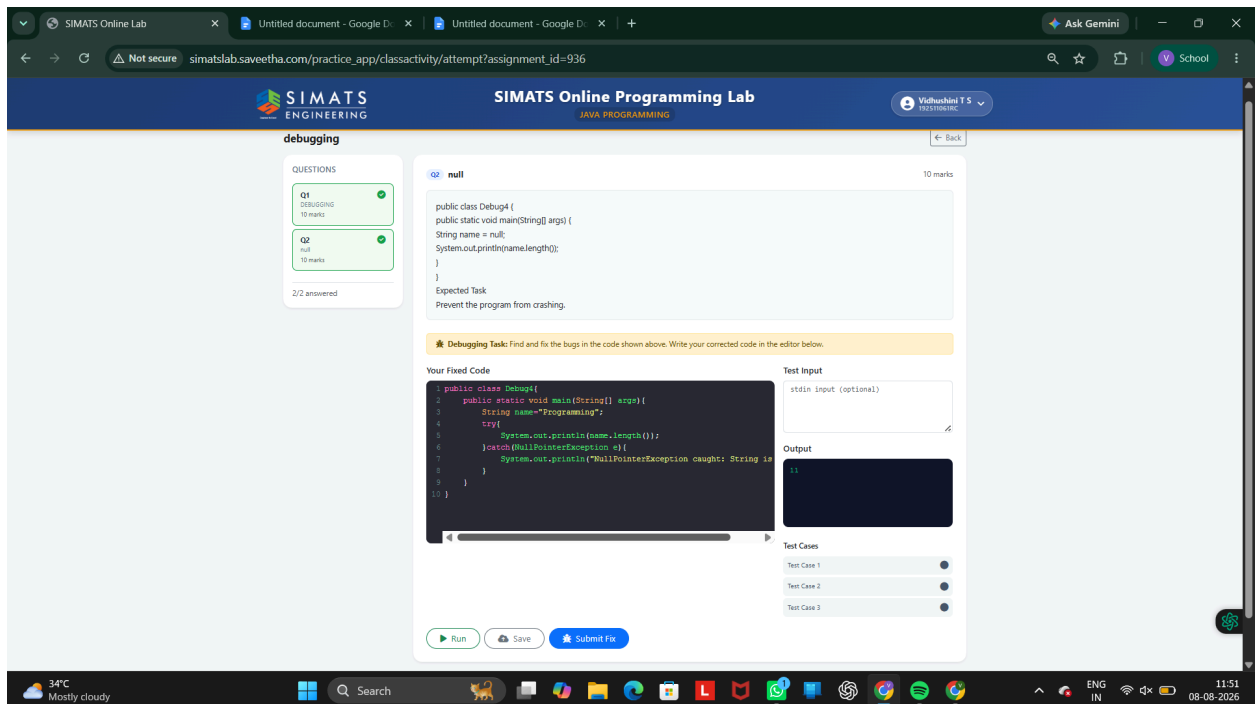
Test Input

10

Output

Division by Zero

2.



The screenshot shows the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" and the sub-header is "JAVA PROGRAMMING". The user is logged in as "Vidhushini T.S". The page is titled "debugging" and shows a "QUESTIONS" sidebar with two questions: "Q1 debugging 10 marks" and "Q2 null 10 marks". The main content area displays a "DEBUGGING" task with a description: "Prevent the program from crashing." The code provided is a Java program that calculates the length of a string. The test input is "Programing", and the output is "11".

```
public class Debug4 {
    public static void main(String[] args) {
        String name = null;
        System.out.println(name.length());
    }
}
```

Expected Task

Prevent the program from crashing.

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
public class Debug4 {
    public static void main(String[] args) {
        String name = "Programing";
        try {
            System.out.println(name.length());
        } catch (NullPointerException e) {
            System.out.println("NullPointerException caught: String is null");
        }
    }
}
```

Test Input

Programing (optional)

Output

11

Test Cases

- Test Case 1
- Test Case 2
- Test Case 3